

Title Placeholder: Physics-Informed Residual Diffusion for OD-Conditioned Trajectory Generation

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Abstract

Accurate simulation of human mobility is fundamental to resilient urban planning and mobility Digital Twins. However, current approaches face a validity gap: deterministic regression models miss stochastic route choice, while generative models can suffer from *macroscopic shrinkage*, producing diverse yet physically unrealistic trajectories with underestimated displacement and speed. We propose **Physics-Informed Residual Diffusion**, which decomposes trajectory generation into a frozen deterministic prior that captures macroscopic motion scale and a diffusion model for stochastic residuals, optionally conditioned on a train-only navigation field. Using a strict dt-fixed ($\Delta t = 30s$), leakage-free protocol on the Shenzhen taxi dataset, we show that residual diffusion substantially improves macroscopic motion statistics while maintaining strong best-of- K coverage. The resulting framework provides a practical path toward simulation-valid, data-driven urban mobility generation.

1 Introduction

Human mobility simulation supports urban analytics and transportation planning, and is increasingly discussed as a key component of mobility “Digital Twins” that enable scenario analysis under uncertainty (Wang et al., 2022). Human mobility is typically goal-oriented; therefore, the origin–destination (OD) endpoints provide essential constraints on feasible future motion. At the same time, even under the same observed history and OD context, multiple plausible future trajectories may exist due to stochastic route choice and local driving variability. Accordingly, a trajectory generator should satisfy two requirements: (i) it should represent this conditional stochasticity, and (ii) it should preserve physical realism at the macroscopic scale, maintaining realistic displacement and speed statistics over the prediction horizon.

These requirements expose a gap in existing approaches. Deterministic regression models (e.g., L2) optimize point-wise error and typically approximate a conditional mean trajectory, which suppresses multi-modality and limits their use for risk-aware analysis. Generative models such as diffusion can produce diverse futures and are often evaluated via best-of- K protocols to assess coverage. However, when applied to city-scale mobility, diffusion samples can exhibit **macroscopic shrinkage**: trajectories systematically under-estimate net displacement and therefore yield biased macroscopic motion statistics, such as mean squared displacement (MSD; displacement growth over time lag) and radius of gyration (Rog; spatial dispersion over a trajectory).

We address the coverage–realism gap with **Physics-Informed Residual Diffusion**. The method decomposes future motion into (i) a frozen deterministic prior that captures OD-driven displacement scale and (ii) a diffusion model that learns stochastic residual deviations around that prior. In addition, we use a train-only navigation field—a non-parametric estimate of local mean flow computed from the training split—as a conditioning signal to improve route-level coverage without leaking information from validation/test data.

Contributions.

- We build a dt-fixed ($\Delta t = 30\text{s}$), leakage-free pipeline, including train-only normalization and navigation-field construction.
- We diagnose macroscopic shrinkage in pure diffusion under a strict evaluation protocol and propose residual diffusion as a structural remedy.
- We report micro-level errors and best-of- K coverage together with macroscopic motion statistics (MSD/Rog) to evaluate simulation validity.

The remainder of the paper reviews related work, states the research question, describes the study region and data, details the methodology, and presents results and discussion.

2 Related Work

2.1 Simulation Validity Requirements in Mobility Digital Twins

Transportation Digital Twins are commonly framed as connected cyber–physical systems that support scenario analysis through continual data integration and a service layer that links physical and digital components (Nag et al., 2025; Yan et al., 2023). In this setting, trajectory generation is not only a forecasting task but also a component of the simulation engine: sampled trajectories serve as inputs to evaluate interventions and risks under uncertainty (Wang et al., 2022; Ferré-Bigorra

et al., 2022). Consequently, model evaluation should go beyond point-wise accuracy and consider *simulation validity*, including (i) stochastic coverage of plausible futures and (ii) physical realism of macroscopic motion statistics.

Traditional mobility simulation relies on Agent-Based Models (ABM) and hand-crafted kinematic rules (Helbing and Molnár, 1995). These approaches are interpretable but often require extensive calibration to match city-scale heterogeneity. Data-driven models provide an alternative by learning mobility patterns directly from large trajectory logs. Early data-driven methods formulated trajectory prediction as deterministic regression using recurrent models or Transformers (Alahi et al., 2016; Zhou et al., 2022; Shi et al., 2022). While effective on mean error, deterministic regression typically produces a single conditional average trajectory, which under-represents multi-modality needed for scenario-level analysis.

Generative sequence models were introduced to recover multi-modality by sampling multiple plausible futures, including VAEs, GANs, and more recently diffusion models (Gupta et al., 2018; Ho et al., 2020; Zhu et al., 2023). For Digital Twin applications, however, diversity alone is insufficient if sampled trajectories are not physically consistent at the macroscopic scale. This motivates methods that explicitly incorporate physical priors and evaluation protocols that jointly measure coverage and macroscopic realism.

2.2 Physics-Informed Inductive Biases in Urban Computing

Physics-Informed Machine Learning (PIML) provides inductive biases to help data-driven mobility models respect urban dynamics. Existing work improves physical realism by incorporating explicit structure, such as graph priors and road-network-constrained generation (Shi et al., 2024; Wei et al., 2024), or by introducing field- and PDE-inspired guidance signals (Bastek et al., 2025).

Our work focuses on *macroscopic flow consistency* for simulation validity. We use a train-only navigation field as an empirical mean-flow conditioning signal and combine it with a residual decomposition so that macroscopic motion scale is anchored by a deterministic prior while diffusion focuses on stochastic deviations.

3 Research Question

We study **OD-conditioned trajectory generation with a known destination** (KnownDestination) under an explicit temporal semantics setting (dt-fixed $\Delta t = 30s$). We investigate the following questions under a strict, leakage-free evaluation protocol:

- **Coverage vs. realism:** under OD conditioning, how do deterministic regression and diffusion-based generators compare in micro-level errors (mean; best-of- K oracle, i.e., minimum error

over K samples) and macro-level motion statistics (net displacement, MSD, Rog)?

- **Main hypothesis:** can a **residual decomposition** (deterministic prior + stochastic residual diffusion) mitigate macroscopic shrinkage, defined as systematic under-estimation of net displacement and speed, while retaining multi-modality?
- **Role of navigation field conditioning:** does injecting a **train-only navigation field conditioning signal** improve best-of- K coverage without introducing conservative bias in macroscopic motion scale?

4 Study Region and Data

4.1 Spatiotemporal Data Curation for Mobility Digital Twins

A mobility Digital Twin is only as reliable as its spatiotemporal data layer. To make *simulation validity* measurable and reproducible, we curate the Shenzhen taxicab GPS dataset (Zhang et al., 2015) into a strict *dt-fixed* benchmark with a leakage-free data contract.

Study region and grid representation. We consider the Shenzhen urban area within a bounding box (latitude [22.45, 22.85], longitude [113.75, 114.65]) and discretize it into a $H_g \times W_g = 400 \times 800$ grid. Trajectory positions are stored as grid coordinates $\mathbf{p}_t = [y_t, x_t]$, which supports city-scale aggregation and the construction of train-only, non-parametric flow statistics.

dt-fixed resampling: why it matters for physics consistency. Raw taxi GPS streams are irregularly sampled, so a “one-step” displacement does not correspond to a fixed physical time interval. This ambiguity directly undermines macroscopic motion statistics (e.g., MSD curves) and any field-based signal that depends on step magnitude. We therefore resample each trajectory to a fixed interval $\Delta t = 30\text{s}$ using linear interpolation, so that each step displacement $\mathbf{v}_t = \mathbf{p}_{t+1} - \mathbf{p}_t$ has an explicit temporal semantics. When physical velocity is needed, it can be obtained by dividing by Δt .

Strict curation rules and dataset scale. We enforce a reproducible resampling contract: timestamps are deduplicated (mean over repeated seconds), monotonicity is required, and any trajectory with a maximum raw gap exceeding 300s is dropped to avoid discontinuous motion and preserve trajectory-level OD semantics. We further remove trajectories whose total duration is shorter than $9\Delta t$ (equivalently, fewer than 10 dt-fixed points). The resulting dt-fixed benchmark spans 2011-04-18 to 2011-04-24. Starting from 154,285 processed trajectories, the benchmark contains

139,634 trajectories, with 10,897 dropped due to large gaps and 3,754 due to insufficient duration. The final split sizes are train=97,222, val=21,103, and test=21,309 trajectories. On the training split, the mean dt-fixed trajectory length is 31.9 points (min=10, max=1,397), corresponding to millions of resampled trajectory points and enabling robust estimation of aggregate statistics. Concretely, the train split alone contains approximately $97,222 \times 31.9 \approx 3.1 \times 10^6$ dt-fixed points.

Leakage-free derived products and OD conditioning. All normalization statistics and the navigation field used for conditioning are computed from the **training split only** and then frozen for validation/test evaluation. This leakage-free protocol is essential for a Digital Twin setting, where conditioning signals must not encode future information. For each trajectory, we define the OD context using the trajectory-level origin and destination endpoints; we then extract sliding windows with observation length $H = 8$ and prediction horizon $F = 12$ under the same dt-fixed semantics.

5 Methodology

This section introduces the OD-conditioned generation task, the compared model families, and the physics-informed residual formulation.

5.1 Problem Setup

Trajectories are represented in a grid coordinate system. Let $\mathbf{p}_t \in \mathbb{R}^2$ denote the 2D position at step t (grid indices in $[y, x]$ order), and define step displacement (velocity) as

$$\mathbf{v}_t = \mathbf{p}_{t+1} - \mathbf{p}_t. \quad (1)$$

With dt-fixed sampling, $\mathbf{v}_t$ has explicit temporal meaning. We denote the observed history as $\mathbf{X} = \{(\mathbf{p}_t, \mathbf{v}_t)\}_{t=-H+1}^0$ and the future velocities as $\mathbf{Y} = \{\mathbf{v}_t\}_{t=1}^F$. Given a global context vector $\mathbf{c}$ (time features and OD endpoints), our goal is to model the conditional distribution

$$p_\theta(\mathbf{Y} \mid \mathbf{X}, \mathbf{c}). \quad (2)$$

In our experiments, we use $H = 8$ observed steps and $F = 12$ future steps under dt-fixed $\Delta t = 30$ s. Given the last observed position $\mathbf{p}_0$, predicted positions are obtained by cumulative summation:

$$\mathbf{p}_k = \mathbf{p}_0 + \sum_{t=1}^k \mathbf{v}_t, \quad k = 1, \dots, F. \quad (3)$$

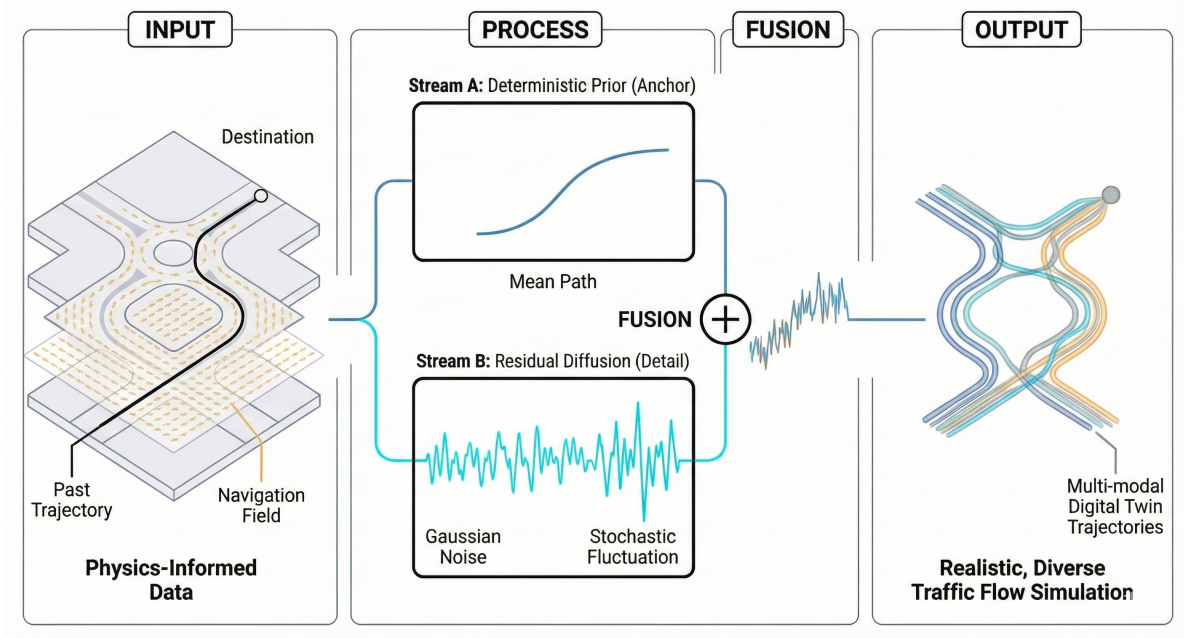


Figure 1: Framework overview. A frozen deterministic prior captures OD-driven macroscopic motion scale. A diffusion model samples stochastic residual deviations, optionally conditioned on a train-only navigation field. The sum yields diverse trajectories while preserving macroscopic motion statistics.

We use *macroscopic shrinkage* to refer to systematic under-estimation of net displacement $\|\mathbf{p}_F - \mathbf{p}_0\|_2$ and mean speed $\frac{1}{F} \sum_{t=1}^F \|\mathbf{v}_t\|_2$, which manifests as biased macroscopic motion statistics such as MSD and Rog.

5.2 Model Framework

We compare a deterministic prior (L2 regression), conditional diffusion generators, and our residual formulation that explicitly separates macroscopic scale from stochastic variability (Figure 1).

Deterministic prior (L2 regression). We train an L2 regression model as a deterministic predictor, and use the same model as the *deterministic prior* in residual variants. The prior predicts a single future velocity sequence, $\mathbf{v}_{1:F}^{\text{prior}} = f_{\text{prior}}(\mathbf{X}, \mathbf{c})$, and is frozen when training diffusion residuals (equivalently $K = 1$ at inference).

Conditional diffusion generator. We adopt a denoising diffusion probabilistic model (DDPM) (Ho et al., 2020) to sample a target sequence conditioned on $(\mathbf{X}, \mathbf{c})$. In the non-residual setting the target is the future velocity sequence; in residual variants the target is the residual velocity sequence

defined below. The model is trained to predict diffusion noise with the standard objective

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{\mathbf{y}_0, t, \epsilon} [\|\epsilon - \epsilon_\theta(\mathbf{y}_t, t, \mathbf{c})\|_2^2], \quad (4)$$

where $\mathbf{y}_0$ is the clean target velocity matrix (future velocities in non-residual variants, residual velocities in residual variants), $\mathbf{y}_t$ is its noisy version at timestep t , and $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

Navigation field conditioning (train-only). To incorporate an empirical flow signal without relying on a road graph, we construct a *navigation field* that summarizes local mean displacement direction and speed in grid space. Importantly, this field is computed using the **training split only** and is fixed during training and evaluation to satisfy the leakage-free data contract. For each grid cell (i, j) we estimate the mean displacement $\bar{\mathbf{v}}_{ij}$ over training steps that fall into the cell, the corresponding mean direction $\mathbf{u}_{ij}$, and mean speed s_{ij} :

$$\bar{\mathbf{v}}_{ij} = \frac{1}{|\mathcal{T}_{ij}|} \sum_{k \in \mathcal{T}_{ij}} \mathbf{v}_k, \quad \mathbf{u}_{ij} = \frac{\bar{\mathbf{v}}_{ij}}{\|\bar{\mathbf{v}}_{ij}\|_2 + \delta}, \quad s_{ij} = \frac{1}{|\mathcal{T}_{ij}|} \sum_{k \in \mathcal{T}_{ij}} \|\mathbf{v}_k\|_2, \quad (5)$$

where $\mathcal{T}_{ij}$ is the set of training steps located in cell (i, j) and δ is a small constant for numerical stability. At inference, we crop a local patch around the last observed position $\mathbf{p}_0$:

$$\Phi(\mathbf{p}_0) \in \mathbb{R}^{3 \times P \times P}, \quad \Phi = [u_y, u_x, s], \quad (6)$$

where u_y and u_x are the two components of the direction field $\mathbf{u} = [u_y, u_x]$. We encode the patch with a CNN encoder and concatenate the resulting embedding with the global condition. In our experiments, we use $P = 32$. We refer to a diffusion generator augmented with navigation field conditioning as *Physics* in our results for brevity.

Main method: Physics-Informed Residual Diffusion. Pure diffusion can exhibit *macroscopic shrinkage* (Eq. 3), where the sampled net displacement scale is systematically smaller than the ground truth. We address this structurally by decomposing future motion into a deterministic prior plus a stochastic residual:

$$\mathbf{v}_{1:F} = \mathbf{v}_{1:F}^{\text{prior}} + \mathbf{v}_{1:F}^{\text{res}}, \quad \mathbf{v}_{1:F}^{\text{prior}} = f_{\text{prior}}(\mathbf{X}, \mathbf{c}), \quad (7)$$

where f_{prior} is the frozen deterministic prior trained under the same dt-fixed protocol. During training, we define the diffusion target as the residual velocity $\mathbf{v}^{\text{res}} = \mathbf{v} - \mathbf{v}^{\text{prior}}$, so the model learns stochastic deviations around the prior rather than the full displacement scale. During inference, we sample $\mathbf{v}^{\text{res}}$ and add it back to the deterministic prior to obtain the final future trajectory.

Destination guidance via classifier-free guidance (CFG). In the KnownDestination setting, destination is a legitimate conditioning input, but its influence can be too weak (leading to conservative motion) or too strong (over-constraining stochasticity). We therefore adopt classifier-free guidance (CFG) as an *inference-time* control knob. During training we randomly drop the destination coordinates in the condition vector with probability p_{drop} to form an “unconditional” branch. At sampling time, we evaluate both conditional and unconditional predictions and combine them as

$$\hat{\mathbf{e}} = \hat{\mathbf{e}}_{\text{u}} + s \cdot (\hat{\mathbf{e}}_{\text{c}} - \hat{\mathbf{e}}_{\text{u}}), \quad (8)$$

where s is the CFG scale, and $\hat{\mathbf{e}}_{\text{c}}/\hat{\mathbf{e}}_{\text{u}}$ denote conditional/unconditional noise predictions. In this essay we treat CFG as a protocolized knob and report two interpretable settings ($s = 2$ and $s = 3$) rather than an open-ended sweep.

5.3 Evaluation Protocol

For generative models we draw K samples per condition and report both mean errors and best-of- K (oracle; minimum over K samples) metrics to quantify coverage. We evaluate micro-level errors using average displacement error (ADE) and final displacement error (FDE) computed on predicted position sequences:

$$\text{ADE} = \frac{1}{F} \sum_{t=1}^F \left\| \mathbf{p}_t - \mathbf{p}_t^{\text{gt}} \right\|_2, \quad \text{FDE} = \left\| \mathbf{p}_F - \mathbf{p}_F^{\text{gt}} \right\|_2. \quad (9)$$

For generative models, mean ADE/FDE average over samples, and best-of- K takes the minimum over the K sampled trajectories for each condition.

We evaluate macro-level physical realism via mean squared displacement (MSD) curves and radius of gyration (Rog). MSD is defined as

$$\text{MSD}(\tau) = \left\langle \left\| \mathbf{p}_{t+\tau} - \mathbf{p}_t \right\|_2^2 \right\rangle, \quad (10)$$

and Rog is computed per trajectory as $\text{Rog} = \sqrt{\frac{1}{F} \sum_{t=1}^F \left\| \mathbf{p}_t - \bar{\mathbf{p}} \right\|_2^2}$, where $\bar{\mathbf{p}}$ is the mean position over the prediction horizon.

Model	K	ADE_{mean}	ADE_{best}	FDE_{mean}	FDE_{best}	MSD_{10}	Rog
Deterministic prior (L2)	1	5.467	5.467	8.855	8.855	304.099	5.494
Diffusion	20	6.741	2.836	11.636	3.950	100.948	3.056
Physics (nav conditioning)	20	6.744	2.510	11.644	3.419	116.869	3.435

Table 1: Diagnostic results on 320 sampled test conditions under $dt\text{-fixed}=30s$. MSD_{10} corresponds to a lag of $10\Delta t = 300s$. Ground-truth macro references on the same conditions are $GT_Rog=5.247$ and $GT_MSD_{10}=349.740$.

6 Results

6.1 Coverage vs. Physical Realism (dt-fixed=30s)

We evaluate under a strict $dt\text{-fixed}$ ($\Delta t = 30s$) protocol. To characterize the coverage–realism tension in non-residual diffusion variants, we first report results on a diagnostic set of 320 sampled test conditions (Table 1). The deterministic prior (L2 regression) is evaluated with $K = 1$, while diffusion-based generators use $K = 20$ samples per condition.

Micro-level errors and coverage. As expected, the deterministic prior achieves the lowest mean errors (Table 1). Diffusion improves multi-modality: best-of- K (oracle; minimum error over K samples) errors are substantially lower, indicating stronger coverage. Navigation field conditioning further improves the oracle upper bound (Physics vs. Diffusion in ADE_{best} and FDE_{best}).

Macro-level physical realism: shrinkage in pure diffusion. Despite strong best-of- K , pure diffusion under-estimates macroscopic motion (Table 1). Navigation field conditioning improves these statistics modestly but does not eliminate the gap. Figure 2 visualizes the same pattern via micro summaries and MSD curves.

6.2 Main Result: Residual Diffusion Mitigates Shrinkage

To separate macroscopic motion scale from stochastic variation, we apply residual diffusion (Eq. 7), using the frozen deterministic prior and learning diffusion residuals. Table 2 reports a leakage-free evaluation on a larger set of 51,200 test conditions ($K = 10$): residual diffusion restores macroscopic scale while retaining strong best-of- K behavior. Residual Physics (residual diffusion with navigation field conditioning) further improves best-of- K , but its macroscopic statistics remain slightly conservative. Here, Speed Ratio is defined as $\mathbb{E}[\|\mathbf{v}\|_2]_{\text{pred}}/\mathbb{E}[\|\mathbf{v}\|_2]_{\text{gt}}$, where $\mathbb{E}[\cdot]$ denotes the mean over future timesteps and evaluated conditions. Rog Ratio is $\text{Rog}_{\text{pred}}/\text{Rog}_{\text{gt}}$, and MSD_{10} Ratio is $MSD_{10,\text{pred}}/MSD_{10,\text{gt}}$ computed on the same conditions.

Model	K	Speed Ratio	Rog Ratio	MSD ₁₀ Ratio	ADE _{best}
Residual Diffusion	10	1.016	1.004	0.935	3.656
Residual Physics (nav conditioning)	10	0.951	0.945	0.864	2.709

Table 2: Residual variants on the test split (51,200 conditions; $K = 10$). Ratios are computed against ground-truth statistics on the same evaluated conditions under the same protocol.

Qualitative geographic-space visualization (Scheme B). To connect model behavior to urban spatial structure, we visualize saved evaluation samples ($N=80$ conditions) in geographic space via a linear bbox projection (without road-level map matching). Figure 3 compares the deterministic prior with residual diffusion. The panel title “Diffusion” denotes the diffusion-based generator; in this figure it corresponds to the residual diffusion variant.

6.3 CFG as an Inference-Time Micro–Macro Knob (Shenzhen Geo-Scale)

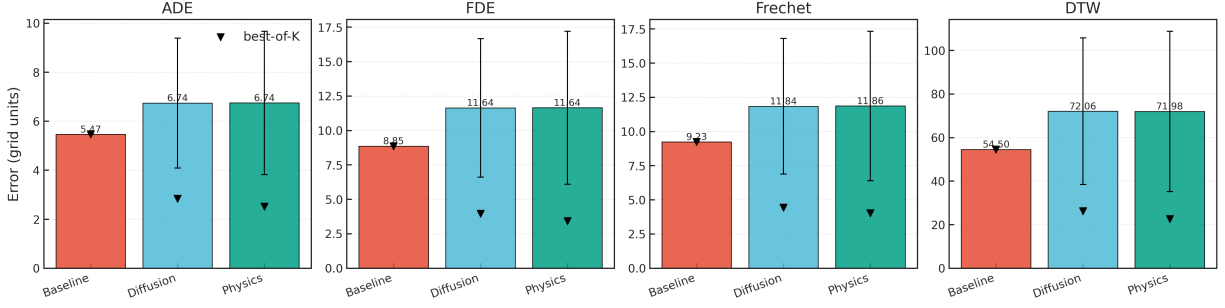
The residual formulation addresses the *structural* shrinkage issue by anchoring macroscopic scale to a deterministic prior. Building on this foundation, we introduce *classifier-free guidance* (CFG) at inference time to control how strongly the generator is attracted to the known destination. This yields an interpretable trade-off: a moderate CFG emphasizes micro-level precision (best-of- K), while a stronger CFG emphasizes macro-level validity (MSD/Rog) by more aggressively enforcing OD intent.

Protocolized settings (avoid hyperparameter swamp). Following the PI review, we report two pre-registered CFG settings rather than an open-ended sweep: **cfg=2** is *micro-optimal* under a macro validity gate, while **cfg=3** demonstrates that the same trained model can be tuned toward *macro-validity* at the cost of micro-precision. Figure 4 summarizes this curve on validation data.

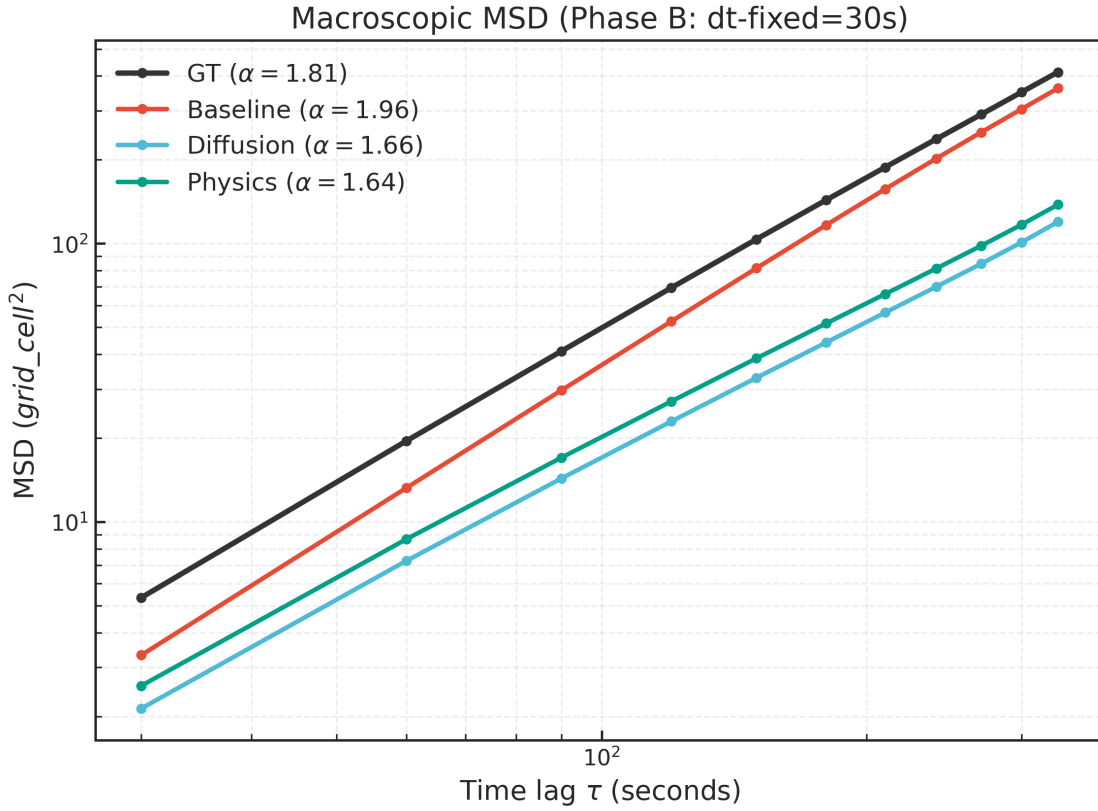
Macroscopic (city-scale) geographic evidence. To link model behavior to Shenzhen’s spatial structure, we project grid trajectories back to latitude/longitude using the dataset bbox (linear mapping) and overlay them on Shenzhen district boundaries (GeoJSON). Although this is not road-level map matching, it provides an intuitive city-scale view of corridor structure and spatial heterogeneity. Figure 5 compares the deterministic prior with CFG2/CFG3: CFG2 preserves the macro scale within the validity gate while keeping a tighter micro fit; CFG3 expands macroscopic dispersion slightly, consistent with stronger destination attraction.

Microscopic (scenario-scale) qualitative cases. Finally, Figure 6 zooms into representative OD conditions and overlays GT, prior, CFG2, and CFG3 on the same local extent. This visualization

highlights *how* the models differ beyond aggregate metrics: CFG2 tends to produce locally smoother, closer-to-GT route shapes, while CFG3 allows larger deviations that increase macroscopic dispersion but can degrade point-wise alignment.



(a) Micro-level metrics (mean/std and best-of- K where applicable).



(b) MSD curves (Pred vs. GT) under dt-fixed=30s.

Figure 2: Quantitative evidence under dt-fixed=30s: diffusion improves best-of- K coverage but exhibits macroscopic shrinkage in MSD/Rog.

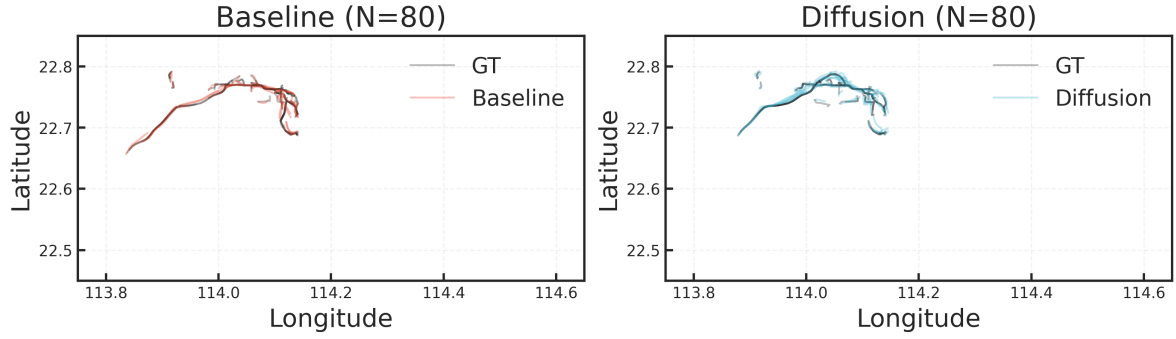


Figure 3: Geographic-space visualization (trajectory overlays). Left: deterministic prior (L2 regression). Right: residual diffusion (v1.1).

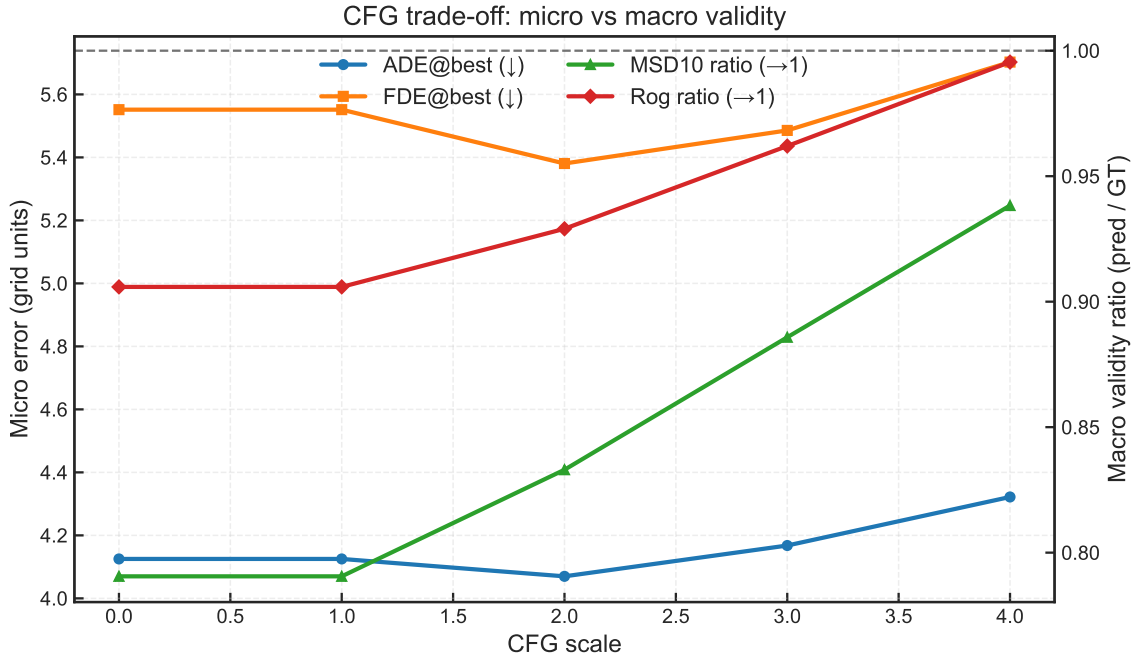
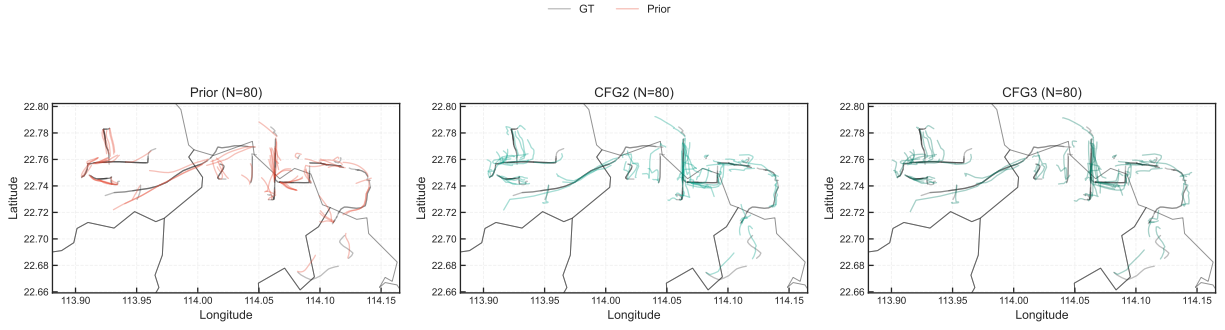
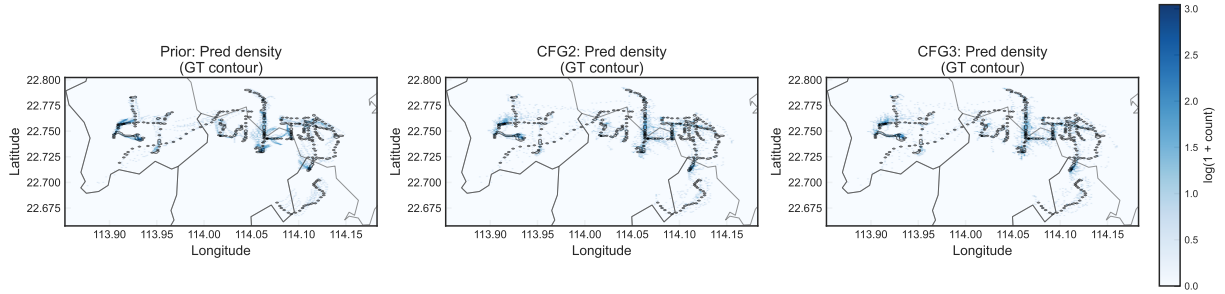


Figure 4: CFG trade-off curve on validation: micro errors (ADE/FDE best-of- K , left axis; lower is better) versus macro validity ratios (MSD_{10}/Rog , right axis; closer to 1 is better). The dashed line at 1.0 indicates the validity gate (pred/GT).



(a) Trajectory overlays in geographic space (saved evaluation subset).



(b) Spatial density of predicted points (heatmap) with GT contour overlay.

Figure 5: Shenzhen geographic-space visualization (no map matching). Caption protocol: *cfg=2 is micro-optimal within macro validity gate; cfg=3 demonstrates the model can be tuned towards macro-validity at the cost of micro-precision.*

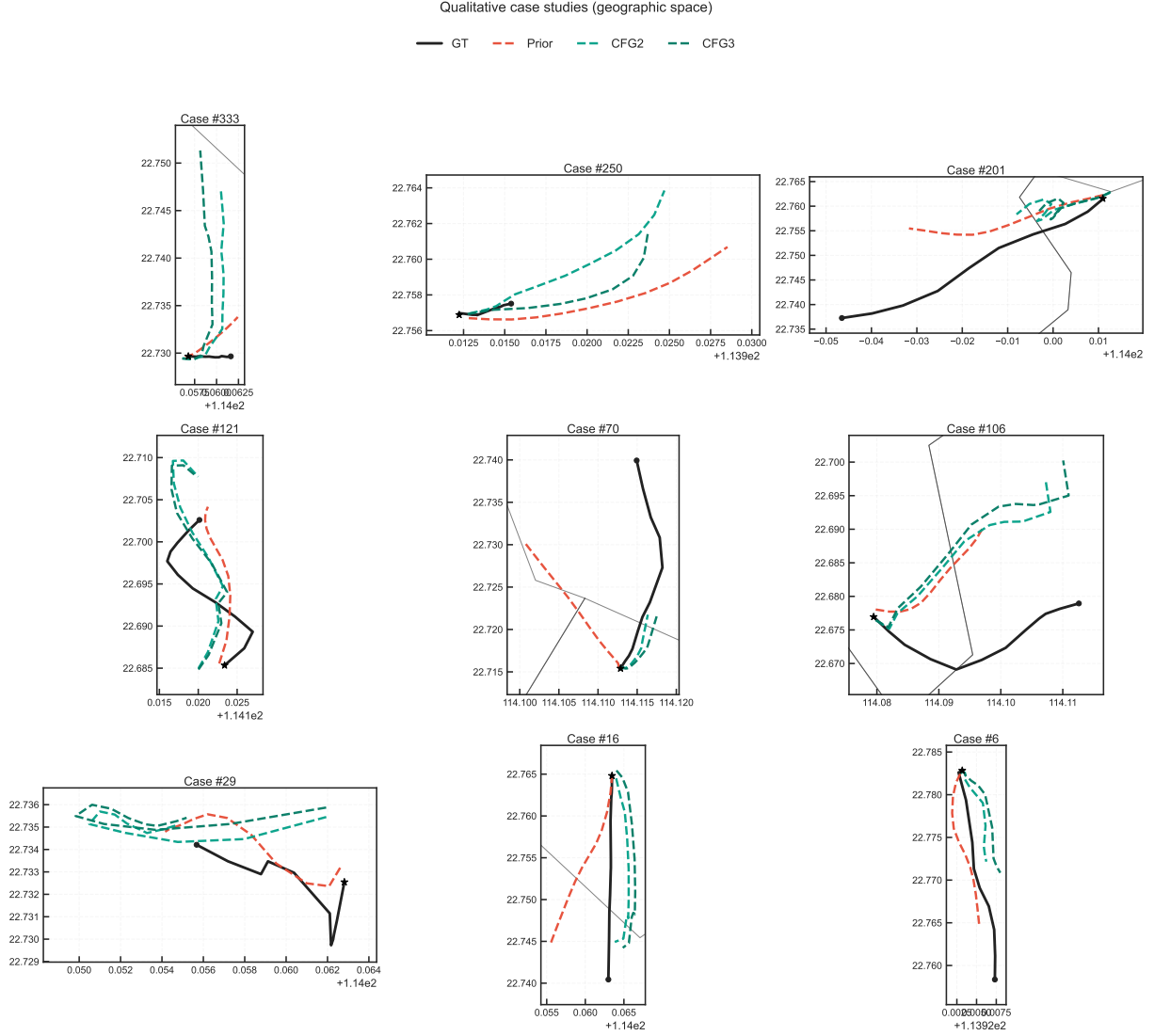


Figure 6: Micro-scale qualitative case studies in geographic space: for the same OD condition, we compare GT, deterministic prior, and two CFG settings ($\text{cfg}=2/3$). Each subplot uses a local zoom extent to highlight route-shape differences.

7 Discussion

Micro–macro tension in generative mobility modeling. Mobility Digital Twins require *simulation validity*: sampled trajectories should cover plausible futures while preserving physically realistic macroscopic motion statistics. Our results show that these objectives are not aligned if one optimizes only point-wise error. The deterministic prior (L2 regression) achieves the lowest mean errors on the diagnostic set, but it produces a single trajectory per condition and therefore cannot represent multi-modality (Table 1). In contrast, diffusion-based generators improve best-of- K coverage, but exhibit *macroscopic shrinkage*: net displacement and speed are under-estimated, leading to lower MSD/Rog (Figure 2).

Effect of navigation field conditioning under a leakage-free protocol. Navigation field conditioning (train-only) improves the oracle upper bound (best-of- K) relative to data-only diffusion (Table 1), indicating that local mean-flow information helps route-level coverage. However, it does not eliminate macroscopic shrinkage. This implies that, under our leakage-free setting, adding a mean-flow conditioning signal alone is insufficient to restore the net displacement and speed scale required for simulation validity. This observation aligns with the intuition that field-based guidance can introduce an implicit bias toward the mean (mean-reversion), dampening the stochastic expansion needed to match true macroscopic dispersion.

Residual diffusion as a structural fix. Residual decomposition separates responsibilities: the deterministic prior carries OD-driven macroscopic motion scale, while diffusion focuses on stochastic residual deviations. This directly targets the shrinkage definition used in this paper (under-estimated net displacement and speed). Empirically, residual diffusion restores macroscopic scale (Speed Ratio and Rog Ratio close to 1) while maintaining strong best-of- K behavior (Table 2).

CFG as a protocolized inference knob. Even after the structural fix, practical digital-twin usage requires *controllability*: planners may prefer higher micro precision for typical scenarios or stronger macro-validity for stress testing and scenario analysis. CFG provides a simple inference-time knob to trade micro and macro objectives without retraining. Crucially, we avoid a parameter swamp by pre-registering two settings (cfg=2 for micro-optimal within a macro gate, cfg=3 for macro-validity emphasis), and we report the full trend only as a transparent validation curve (Figure 4).

Limitations and future work. Our current pipeline is grid-based. For geographic presentation we use a linear bbox projection to latitude/longitude and overlay Shenzhen district boundaries from GeoJSON; this improves interpretability but is not road-level map matching. Future work should incorporate road-graph structure and explore more expressive conditioning mechanisms (e.g.,

ControlNet-style injection or learnable gating) so that navigation field information guides direction without acting as an overly restrictive constraint. Finally, stronger conclusions require multi-seed replication and broader evaluation under the same strict data contract.

8 Conclusion and Future Work

We presented a leakage-free, dt-fixed benchmark for OD-conditioned trajectory generation and demonstrated a key limitation of pure diffusion models: *macroscopic shrinkage*, where samples improve best-of- K coverage yet under-estimate displacement and speed. To address this, we proposed **Physics-Informed Residual Diffusion**, which anchors macroscopic motion scale with a frozen deterministic prior and lets diffusion model only stochastic residuals, optionally conditioned on a train-only navigation field.

Future work should (i) incorporate road-graph constraints beyond grid representations, (ii) design adaptive mechanisms so that mean-flow conditioning signals improve coverage without overly constraining macroscopic motion, and (iii) validate stability through larger-scale, multi-seed evaluations under the same strict data contract. Ultimately, this framework provides a reliable, physics-consistent foundation for next-generation mobility Digital Twins, enabling planners to simulate diverse future scenarios without sacrificing macroscopic realism.

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